Accenture — Aptitude Test 1

Questions 1–25 of 100 · 25 questions

Question 1. What is the next number of the following sequence 2,2,12,12,30,30,_?

Solution:
2 = 11 + 1
2 = 22 – 2
12 = 32 + 3
12 = 42 – 4
30 = 52 + 5
30 = 62 – 6
7th term = 72 + 7 = 56

Question 2. What is the missing number in this series 8, 2, 14, 6, 11, ___ , 14, 6, 18, 12?

Solution:
Sum of first digit and last digit = 20
Sum of second digit and second last digit
= 20
Sum of third digit and third last digit
= 20
Similarly, the sum of 11 and the missing
number = 20
11 + X = 20
X = 9
The missing number is 9.

Question 3. Find the missing number in the series 15, 51, 216, 1100, ___ ,46452?

Solution:
15 x 3 + 6 = 51
51 x 4 + 12 = 216
216 x 5 + 20 = 1100
1100 x 6 + 30 = 6630
6630 x 7 + 42 = 46452
The missing number is 6630.

Question 4. What is the next number of the following sequence 2, 12, 36, 80, 150,.... ?

Solution:
13 + 12 = 2
23 + 22 = 12
33 + 32 = 36
43 + 42 = 80
53 + 52 = 150
So the missing number is
63 + 62 = 216 + 36
= 252.

Question 5. What is the 56743rd term in the series 1234567891011121314…….?

Solution:
Number of 1 digit numbers = 9
Number of 2 digit numbers = 90
Number of 3 digit numbers = 900
Number of 4 digit numbers = 9000
So, the numbers of terms till the last 4 digit number
= 9x1 + 90x2 + 900x3 + 9000 x 4
= 9 + 180 + 2700 + 36000
= 38889
56743 – 38889 = 17854
For the remaining 17854 digits, there are groups of 5 digit numbers.
17854/5 = 3570 and reminder 4
Among the 3590 set of 5 digit numbers, the first number is 10000 and the last number is 13569
So, 56739th term = 9 and
56740th term to 56743rd term = 1357
56743rd term = 7

Question 6. One dog tells the other that there are two dogs in front of me. The other one also shouts that he too had two behind him. How many are there?

Solution:
There are three dogs.
The three dogs are walking in a circular pattern making both statement true.

Question 7. A number of cats got together and decided to kill between the 999919 mice. Every cat killed an equal number of mice. Each cat killed more mice than there were cats. How many cats do you think there were?

Solution:
Assume that the number of cats = x and number of mice killed by each cat = y
xy = 999919
999919 = 100000 – 81
= (1000)2 – (9)2 = (1000 + 9)(1000-9)
= 1009 x 991
Number of cats
=991

A c c e n t u r e

Question 8. 70 students are required to paint a picture. 52 use green color and some children use red, 38 students use both

the colors. How many students use red color?

Question 9. In a class total number of students is 15 in which 7 students speak English and 8 student speak Hindi, 3 students can not speak in both languages. So how many student can speak both languages?

Solution:

3

7
English

8
Hindi

Question 10. There are 76 people residing in a colony. 53 read Hindu, 46 read Times, 39 read Deccan and 15 read all. If 22 read Hindu and Deccan, 23 read Deccan and Times, then what is the number of persons who reads only Times and Hindu?

Solution:

Solution:

Red
Green

38

x

52

Number of students using green alone
= 52 – 38 = 14
Number of students using green or red and green = 14 +38 = 52
The number of students using red alone
= 70 – 52 = 18
Number of students using red = 18 + 32
= 50

Number of people who read only Deccan and Hindu = 22 – 15 = 7
Number of people who read only Times and Deccan = 23 – 15 = 8
Number of people who read only Deccan
= 39 – 7 – 8 – 15 = 9
Number of people who read only Hindu
= 53 – 7 – 15 – x
Number of people who read only Times
= 46 – 8 – 15 – x
Total people

= 15 + 7+ 8 + x + 9 + (53 – 7 – 8 – x) +
(46 – 8 – 15 – x) = 76
100 – x = 76
x = 24
Number of people who read both Hindu
and Times = 24

Question 11. At an international conference, 100 delegates spoke English, 40 spoke French, and 20 spoke both English and French. How many delegates could speak at least one of these two languages?

A c c e n t u r e

15

Number of students who speak at least
one language = 15 – 3 = 12
P(A■B) = P(A) + P(B) – P(A■B)
Number of students who speak both the
languages
= 7 + 8 – 12
= 3

Solution:
Number of people who speak English
alone = 100 – 20 = 80

Hindu
Times

x

53
46

15

23
22

39

Deccan

Number of people who speak French alone = 40 – 20 = 20
Number of people who speak at least one language = 100 + 20 + 20 = 140.

Question 12. The sum of 3 single digit numbers is 15 less than their product. If we subtract 2 from first given number then sum of these numbers will become 7 more than their product. The product of given 3 numbers will be?

Solution:
Assume that the tree 1 digit numbers are x, y and z
Sum of the three numbers = x + y + z
Product of the three numbers = xyz.
Given,
xyz – (x + y + z) = 15 ---(1)
And,
[(x-2) + y + z] – [(x-2)yz] = 7
(x + y + z) – xyz + 2yz = 9 ---(2)
(1) + (2) will give
2yz = 24
yz = 12
There are two solutions for y and z. They are:
(y, z) = (3, 4) or (2, 6)
Substitute (3, 4) in (1), we get
12x – x – 3 – 4 = 15
11x = 22
x = 2
Substitute (2,6) in (1) we get
12x – x – 2 – 6 = 15
11x = 23
x = 2.09
We know that x is a 1 digit number. So the only solution for (y, z) = (3, 4) and

x = 2
Product of three numbers = 2x3x4 = 24

Question 13. If N is the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. What is the sum of the digits of N?

Solution:
N = HCF of difference between the numbers
Difference between the numbers is

= 3360, 2240, 5600
To find HCF (3360, 2240, 5600), first take the prime factors of each number.

Prime factors of 3360
= 2x2x2x2x2x5x7x3 = 25 x 51 x 71 x 31

Prime factors of 2240
= 2x2x2x2x2x2x7x5 = 26 x 51 x 71

Prime factors of 5600
= 2x2x2x2x2x5x5x7 = 25 x 52 x 71

HCF of prime factors = product of common factors raised to their least powers.
So, HCF(3360, 2240, 5600)
= 25 x 51 x 71 = 1120
The number 1120 is the greatest number which will divide 1305, 4665 and 6905 and leave the same reminder.
N = 1120
Sum of the digits of N = 1+1+2+0 = 4.

Question 14. Find the numbers between 100 and 400 which is divisible by either 2,3,5,7?

Solution:
Number of terms divisible by 2 or 3 or 5 or 7 = Number of terms divisible by 2 + Number of terms divisible by 3 + Number of terms divisible by 5 + Number of terms divisible by 7 –
Number of terms divisible by 2 and 3 –
Number of terms divisible by 2 and 5 –
Number of terms divisible by 2 and 7 –
Number of terms divisible by 3 and 5 –
Number of terms divisible by 3 and 7 –
Number of terms divisible by 5 and 7 –
Number of terms divisible by 2, 3, 5 –
Number of terms divisible by 2, 3, 7 –
Number of terms divisible by 3, 5, 7 –
number of terms divisible by 2, 5, 7 –
Number of terms divisible by 2,3,5,7
Using the formula to find the number of terms in an arithmetic progression we can find the number of terms in each category.
The formula is:
N = [(l – a)/d] + 1

Where l = last term of the series
a = first term of the series
d = common difference between terms
N = number of terms.

A c c e n t u r e

N2 = [(398 – 102)/2] + 1 = 149
N3 = [(399 – 102)/3] + 1 = 100
N5 = [(395 – 105)/5] + 1 = 59
N7 = [(399 – 105)/7] + 1 = 43
N2,3 = [(396 – 102)/6] + 1 = 50
N2,5 = [(390 – 110)/10] + 1 = 29
N2,7 = [(392 – 112)/14] + 1 = 21
N3,5 = [(390 – 105)/15] + 1 = 20
N3,7 = [(399 – 105)/21] + 1 = 15
N5,7 = [(385 – 105)/35] + 1 = 8
N2,3,5 = [(390 – 120)/30] + 1 = 10
N2,5,7 = [(350 – 140)/70] + 1 = 4
N2,3,7 = [(378 – 126)/42] + 1 = 7
N3,5,7 = [(315 – 105)/105] + 1 = 3
N2,3,5,7 = [(210 – 210)/210] + 1 = 1
The required answer is:
149 + 100 + 59 + 43 – 50 – 29 – 21 – 20
– 15 – 8 – 10 – 4 – 7 – 3 – 1 = 183

Let the number be X
Given,
(1/3)(1/4) X = 15
X = 15 x 3 x 4 = 180
One – tenth of the number is
= (1/10) x 180 = 18.

Question 15. A number consists of 3 digits whose sum is 10. The middle digit is equal to the sum of the other two and the number will be increased by 99 if its digits are reversed. The number is?

Solution:
LCM of 26, 39, 52, 65 is the smallest
number which will leave reminder = 0
LCM(26, 39, 52, 65) = 780
The difference between the divisor and
the reminder in each case = 4.
So subtract 4 from 780.
The required number = 780 – 4 = 776

Solution:
Let the three digit number be xyz which
can be written as 100x + 10y + z
When the number is reversed we get zyx
which can be written as 100z + 10y + x
Given y = x + z
So,
100x + 10(x + z) + z = number xyz
100z + 10(x + z) + x = number zyx
From these two equations, we get

110z + 11x – 110x – 11z = 99
99z – 99x = 99
z – x = 1
(x, y, z) can have different values:
(x, y, z) = (1, 3, 2) or (2, 5, 3) or (3, 7, 4)
and so on.
But only for the values (2, 5, 3), the sum
of the digits = 10
The number is 253.

Question 16. If one-third of one-fourth of a number is 15, then three-tenth of that number is?

Solution:

Solution:

The two digits will be either (1, 8) or
(2, 4) because the product of them is 8.
If the number is 18, when reversed it
becomes 81, which is not 18 more than
18.
But if the number is 24, when it is
reversed, we get 42 which is 18 more
than 24.
The required number is 24.

Question 17. The ratio of two numbers is 3:4 and their HCF is 4.Their LCM is?

Solution:
Let us assume that the two numbers are
3x and 4x. Then x will be the HCF.
HCF = x = 4
The two numbers are 12 and 16.
LCM of 12, 16 = 48

Question 18. Find the smallest number which leaves 22,35, 48 and 61 as remainders when divided by 26, 39, 52 and 65 respectively.

Question 19. The sum of the squares of three numbers is 138, while the sum of their products taken two at a time is 131. Their sum is?

Solution:
Le the three numbers be x, y, z
Given that x2 + y2 + z2 = 138
xy + yz + xz = 131
(x+y+z)2 = x2+y2+z2+ 2(xy + yz + xz)
(x+y+z)2 = 138 + 2(131) = 400

x+y+z = 20
Sum of the three numbers = 20.

A c c e n t u r e

Question 20. A two-digit number is such that the product of the digits is 8. When 18 is added to the number, then the digits are reversed. The number is?

Question 21. What is the sum of two consecutive even numbers, whose squares give the difference 84?

Solution:
Let the first be x.
Then the second number is x + 2.
Given that
(x+2)2 – x2 = x2 + 4x + 4 – x2 = 84
4x = 80
x = 20
The first number is 20 and the second number is 22
Sum of the numbers = 42.

Question 22. 215 : 474 :: 537 : ___?
a) 26
b) 27
c) 25
d)22

Solution:
2+1+5+4+7+4 = 23
5+3+7+x = 23
x = 8
When the digits of the missing number is added, we must get 8.
Only from option (a) 26, we get 2 + 6 =8

Question 23. A 20 liter mixture contains milk and water in the ratio 3:5. If 4 liters of mixture is replaced with 4 liters of water what is the final ratio of milk and water?

Solution:
After removing 4 liters, we will be left with 16 liters.
Quantity of milk = (3/8) x 16 = 6
Quantity of water = (5/8) x 16 = 10
When 4 liters of water is added we will have 6 liters of milk and 14 liters of water.
Final ratio of Milk and Water = 6 : 14

Question 24. If the ratio of work done by (x-1) men in (x+1)days to the work done by (x+2)men in (x-1)days is 9:10, then x is equal to?

Solution:
Work done is directly proportional to number of men and the time taken.
From the question, we can derive
W1 : W2 = (x-1)(x+1) : (x+2)(x-1)
9/10 = (x-1)(x+1) : (x+2)(x-1)
9/10 = (x+1)/(x+2)
9x + 18 = 10x + 10
x = 8

Question 25. In a theatre the ratio of car & two wheeler is 1:8. Total Number of tyres are 100. Find how many are cars and how many are two wheelers?

Solution:
Let the number of cars = x and
Let the number of two wheelers = 8x
Number of car wheels = 4x
Number of bike wheels = 2(8x) = 16x
4x + 16x = 100
x = 5
Number of cars = x = 5
Number of bikes = 8x = 8(5) = 40